2024 AMC 10B Problem 10 - Solution

Review the full statement and step-by-step solution for 2024 AMC 10B Problem 10. Great practice for AMC 10, AMC 12, AIME, and other math contests

2024 AMC 10B Problem 10

Problem:

Quadrilateral $ABCD$ is a parallelogram, and E is the midpoint of the side $\overline{AD}$. Let F be the intersection of lines EB and AC. What is the ratio of the area of quadrilateral $CDEF$ to the area of $\triangle CFB$?

Answer Choices:

A. $5:4$

B. $4:3$

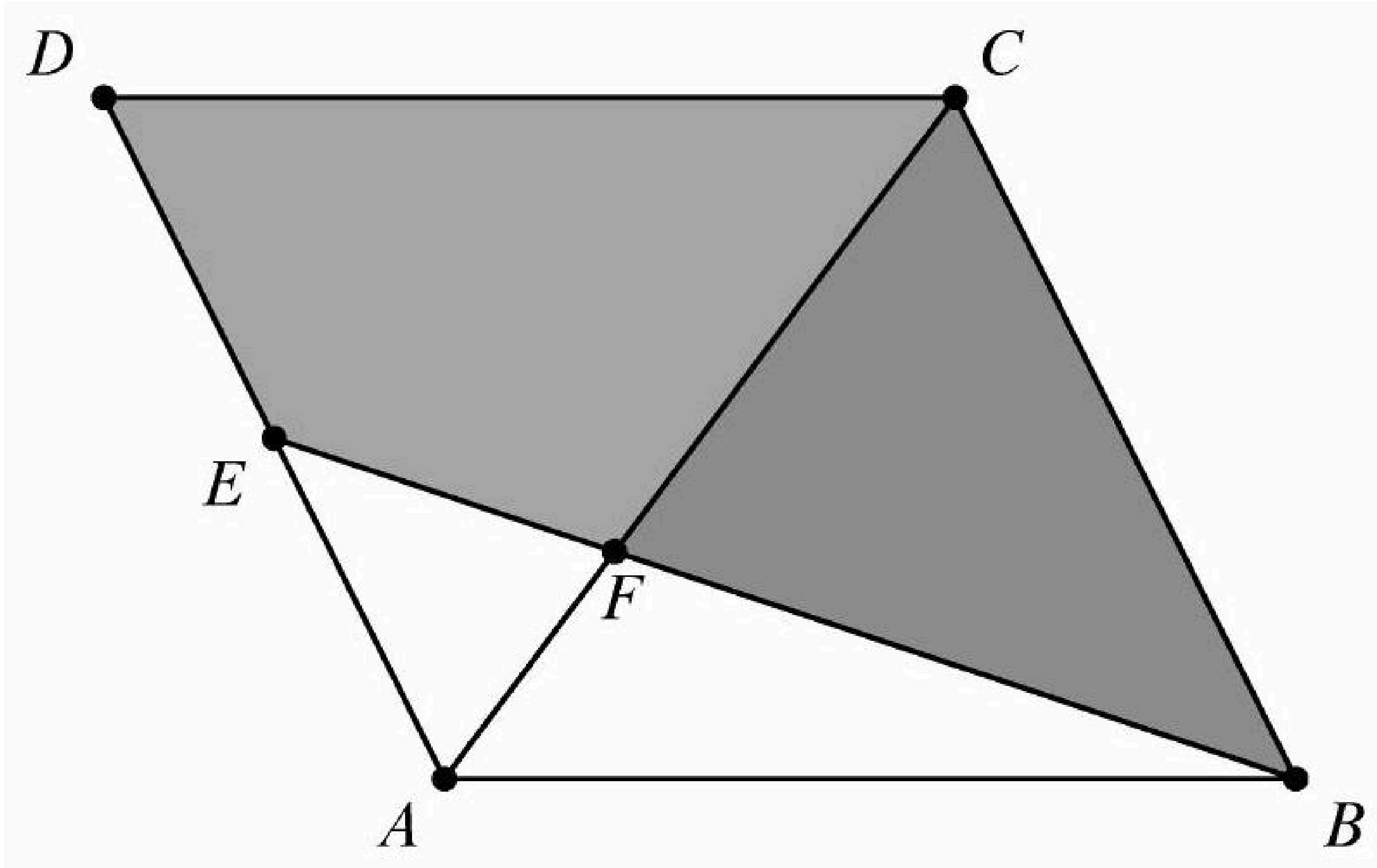
C. $3:2$

D. $5:3$

E. $2:1$

Solution:

Triangles $\triangle AFE$ and $\triangle CFB$ are similar by Angle-Angle. The ratio of corresponding sides is $1:2$, so the ratio of their areas is $1:4$. Furthermore, consider $\triangle AFE$ and $\triangle AFB$. Because $FE : FB = 1:2$ and the heights of the two triangles corresponding to bases $\overline{FE}$ and $\overline{FB}$ are the same, their areas are in a $1:2$ ratio. Finally, observe that the area of $\triangle ABE$ is $\frac{1}{4}$ the area of parallelogram $ABCD$. Therefore, using the area of $\triangle AFE$ as unit, the area of $\triangle CFB$ is 4 , the area of $\triangle AFB$ is 2 , the area of quadrilateral $ABCD$ is $4 \cdot (1+2) = 12$, and the area of quadrilateral $CDEF$ is $12 - (1+4+2) = 5$. The requested ratio of the areas of quadrilateral $CDEF$ and $\triangle CFB$ is $\text{(A)}\boxed{5:4}$.



Note: The answer can also be obtained by assuming that the parallelogram is a square with vertices $(0, 0)$, $(12, 0)$, $(12, 12)$, and $(0, 12)$.